Inverted MAML Card 1: Round 7 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.035, L2=768, Mom=0.9, Clamp=±75 LSB

1. Round 7 Summary & Cumulative Comparison (R1–R7)

• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 7 Detrended Trend Metrics:** Lift vs E_base = **+0.18 b**, Loss Ratio = **1.38x**, Climb = **+0.48 b**, Jitter = **0.26 b**, Recovery = **0.95**, Score = **+1.191**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R1	0.03	1024	0.9	±75	+0.66b	1.57x	+0.74b	0.33b	0.67	+1.957
R2*	0.035	1536	0.85	±75	+0.50b	1.38x	+0.82b	0.29b	0.96	+2.004
R3	0.02	2560	0.8	±70	+0.09b	1.11x	+0.22b	0.26b	1.00	+0.559
R4	0.045	2048	0.88	±75	+0.11b	1.17x	+0.14b	0.24b	1.00	+0.643
R5	0.03	1024	0.85	±75	+0.20b	1.34x	+0.43b	0.26b	0.89	+1.147
R6	0.025	1280	0.92	±75	+0.06b	1.12x	+0.34b	0.23b	1.00	+0.689
R7	0.035	768	0.9	±75	+0.18b	1.38x	+0.48b	0.26b	0.95	+1.191

2. Phase-by-Phase Telemetry (3,917 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	334	9.38->9.42 (9.42b)	-	4438->4204	[27,18]	1	333
P2	Cold / Regler OFF	296	9.40->9.38 (9.39b)	-	5993->4965	[0,0]	0	0
P3	Warm / Regler ON	1066	9.39->9.35 (9.38b)	+0.01 b	5141->4725	[34,33]	3	1066
P4	Warm / Regler OFF	933	9.31->8.83 (8.99b)	+0.02 b	5220->11143	[0,0]	0	0
P5	Warm / Regler ON	817	8.74->8.76 (8.80b)	+0.29 b	11691->12402	[75,75]	19	817
P6	Warm / Regler OFF	471	8.42->8.31 (8.35b)	+0.03 b	18601->22121	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 7

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 7 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

